

2(a)	extracellular (enzyme) ; R exocellular	1
2(b)(i)	arrow between O and CO on lower fatty acid residue ;	1
2(b)(ii)	hydrolysis ; ester ;	2

Question	Answer	Marks
2(c)	<u>substrate / olive oil</u>, concentration ; for, lipase / the enzyme, to achieve, <u>half</u>, maximum rate (of reaction) / $V_{\max}$ (at a temperature of 37 °C) ; R K_m is half, maximum rate of reaction / $V_{\max}$ <i>ref. to lipase / enzyme not needed if the context is correct</i> ‘when the rate of reaction is half $V_{\max}$, the olive oil concentration is 91.76 mmol dm⁻³’ <i>is two marks</i>	2

Question	Answer	Marks
2(d)(i)	any four from: accept activity for percentage enzyme activity throughout check for correct units – must be seen once <u>optimum</u> temperature is 80 °C ; - increasing temperature increases activity, up to, 80 °C / the optimum / the maximum ; I constant increase I enzyme more active R if implied a continued increase throughout the temperature range if mp not gained, allow extracted values: from 25 °C to 80 °C, percentage / activity, from 57 to 100 - increased kinetic energy (of, substrate / enzyme, molecules) ; A increased energy if movement stated - increase, in enzyme-substrate collisions / AW, (per unit time) ; A increase in number of enzyme-substrate complexes must be in context of only up to 80 °C allow ecf if mp2 rejected - above, optimum / 80 °C, decrease in activity ; ecf for incorrect optimum A percentage activity, from 100 to 55 / 56 for decrease in activity - partial (progressive) denaturation ; A enzyme begins to denature A progressive denaturation ora enzyme not completely denatured - explanation of denaturation / partial denaturation ; e.g. loss of, tertiary structure / shape of active site breakage of bonds between R-group breakage of two named bonds A fewer enzyme-substrate collisions if mp4 not gained - reason for, high optimum temperature / activity high at higher temperature ; e.g. is a thermostable enzyme enzyme has more bonds R more peptide bonds enzyme more compact extracted from bacteria living in hot springs enzyme protected by other proteins	4

Question	Answer	Marks
2(d)(ii)	<i>any two from:</i> active over a (much) narrower range of temperature ; optimum of / maximum activity at / peaks at, 35 °C to 40 °C ; steeper increase (from 25 °C) to optimum / maximum activity / peak <u>and</u> steeper decrease after optimum ; decreases to, zero / 0, (percentage) activity (after optimum) ; A becomes denatured when temperature higher than (approx. after) 40 °C I will be 0 percentage activity at 120	2

Question	Answer	Marks
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